

*BB Scattering: Writing (Part 2)***Feedback on figures**

You will be assigned 3 figures from another team. Follow the instructions on Canvas to submit comments that will help the other team improve their figures. **Make sure to do this before continuing!**

Mapping out your report

Before you start to write in earnest, you should plan the document with your teammates. This will make writing easier and help the text be more coherent.

1. Start a document for your report. It doesn't matter what you use at this point as long as all the team members have concurrent access to it (Google Docs, Overleaf, etc.).
2. Create section titles for your report:
 - Title page including an abstract
 - Introduction
 - Theory
 - Methods
 - Data and analysis
 - Conclusion
 - Team Roles and Acknowledgements
 - References
3. Insert your figures to the appropriate sections. Note that you may wish to replace your current versions of the figures once you receive feedback on them, but you can still put them in place to plan around them.
4. Review the instructions for each of the sections above in the “Guide to writing a lab report” linked to Canvas. Then go through each section of your lab report and create some bullet points identifying the main things you want to include and the main points you want to make. These ideas can definitely change as you write your report and your thoughts become clearer! Make sure that the narrative of your results across the whole report is represented in your outline. *Note: your final report will not include these bulleted lists—as you write the full version of your report, you will replace the bullets with paragraphs.*

Once you have some notes in each section, have each member of your team choose a section to start filling in more detail. Some ideas for getting started are given below:

Starting your Introduction

Review the section in the “Guide to write a lab report” about introductions. If you have not already done so, do some research on about scattering experiments in physics (see instructions in the previous writing assignment). Add bullets or sentences summarizing some of these experiments and how our BB scattering experiment relates to the physics of those other experiments. At the same time, add references to the external sources where you found this information in your References section.

Starting your Methods

Review the section in the “Guide to write a lab report” about Methods. Write down the steps by which you collected the data, including references to your experimental setup figure. Write as if the reader had not already done the experiment themselves and did not have access to the instructions that you followed. Also make sure to include any specific choices you made that might the precision and accuracy of your result—you may wish to review the Data Acquisition assignment you completed where you described any such choices. You may also reference the instructions provided for the data acquisition, but make sure to describe your methods *in your own words!*

Starting your Theory

Review the section in the “Guide to write a lab report” about Theory. Motivate the mathematical expressions you will use for your analysis and define the variables you will be using. If you have a figure that relates to this section, explicitly describe how the equations are related to the figure and the experiment. You should have math here!

Starting your Data and Analysis

Review the section in the “Guide to write a lab report” about Data and Analysis. Describe your data and analysis figures, including the manner in which you processed the data raw data into your final figure(s), their uncertainties, and how you performed any fits. Describe your final radius measurement (including uncertainty) and compare it to a physical measurement of the radius, and interpret the agreement or disagreement between those values.

Planning the rest of the work

Before leaving for the day, make a plan with your colleagues about who will do what with the rest of the report and when that work is due. You should leave time for edits when the document is fully written. Many lab reports read as a collection of text written by different people, so try to avoid this pitfall.

When you are done for today, upload your outline (and any additional text you have filled in) to Canvas as a PDF.